Tarek Hamid

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RESEARCH INTERESTS

My research develops **source-aware wearable sensing systems** that combine richer physiological measurements with contextual evidence to enable robust, personalized health monitoring. I jointly design sensing devices and computational models to understand how physiology, interactions between the sensor and body, and environmental conditions contribute to measured signals. My research spans three complementary directions: **(1) information-rich wearable devices**, expanding sensing capabilities through approaches such as multiwavelength optical sensing to reveal physiological information inaccessible to conventional configurations; **(2) contextual sensor integration**, combining measurements across wearable form factors, instrumented everyday objects, and environmental sensors to provide evidence beyond a single device; and **(3) physics-informed signal decomposition**, using these richer, contextualized measurements to distinguish underlying sources of variation and develop models that account for individual physiology. Together, these directions aim to expand what wearables can measure and improve how those measurements can be interpreted across people and real-world conditions.

EDUCATION

University of Virginia, Charlottesville, VA *Jan 2024 - Present*
Ph.D. in Electrical and Computer Engineering
Dissertation: *Broad-Spectrum Multi-Wavelength PPG Algorithms for Personalized and Robust Wearable Physiological Monitoring*
Advisor: Prof. Amanda Watson

University of Pennsylvania, Philadelphia, PA *Dec 2023*
M.S. in Computer Science (50% Completed)

Johns Hopkins University, Baltimore, MD *Dec 2019*
M.S. in Biomedical Engineering

The College of New Jersey, Ewing, NJ *May 2017*
B.S. in Biomedical Engineering

HONORS AND AWARDS

Google PhD Fellowship Finalist *Apr 2026*
National finalist for the Google PhD Fellowship ($170,000 support over two years)

Runner-up for Best Paper: IEEE Pervasive Computing *Apr 2026*
Characterization and Feasibility of Wearable Spectroscopic Tracking of Nutrition Biomarkers was selected as runner-up for best paper of IEEE Pervasive Computing 2025.

UVA Research Excellence Award Finalist *Apr 2026*
Finalist for UVA Engineering Award for Research Excellence.

Best Talk: SandboxAQ 2026 Conference *Mar 2026*
Award of $300 for best talk at the 2026 SandboxAQ Conference.

SandboxAQ Global Travel Award *Aug 2025*
Award of $5,000 to cover travel expenses to present at UbiComp'2025 in Helsinki, Finland.

UVA Precision Health Initiative Grant *Feb 2025*
Award of $4,000 to cover travel expenses to present at ICASSP'25 in Hyderabad, India.

Runner-up for Best Paper Award, IEEE BSN 2024 *Oct 2024*
Presented Raproto: An Open-Source Platform for Rapid Prototyping with Wearable Devices at IEEE Body Sensor Networks 2024.

TCNJ Merit Scholarship *Aug 2013 - May 2017*
Award of $32,000 for educational expenses.

PUBLICATIONS

1. *DermaGlow: Objective Quantification of Melanin, Erythema and Skin-tone Using Wearable Optical Spectroscopy*
   Tarek Hamid, Patricia Flores, Jane Byun, Elizabeth C. Courtney, Kyle C. Quinn, Amanda Watson
   Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT).
   US Patent #10885-083US1

2. *Characterization and Feasibility of Wearable Spectroscopic Tracking of Nutrition Biomarkers*
   Tarek Hamid, Elizabeth Courtney, Patricia Flores, Jane Byun, Afsaneh Doryab, Sibylle Kranz, Amanda Watson
   Pervasive'25: IEEE Pervasive Computing: Special Issue on Bio Sensing (Featured Paper)

3. *A Multi-Wavelength Optical Sensing Framework for Calibration-Free Wearable Blood Pressure Monitoring*
   Tarek Hamid, Patricia Flores, Jane Byun, Xi Chen, Haoran Zhang, Kyle Quinn, Amanda Watson
   International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2025.
   US Patent #10885-090US1

4. *Raproto: An Open-Source Platform for Rapid Prototyping with Wearable Devices*
   Tarek Hamid, Kimberly Helm, Hyonyoung Choi, Jean Park, Claire Kendell, Stephanie Cummings, Steve Messe, Stefanie Modri, Insup Lee, Amanda Watson, James Weimer
   Proc. IEEE-EMBS Int. Conf. on Body Sensor Networks (IEEE BSN), 2024.

5. *Wearable Sensing for Measuring Skin-tone, Melanin, and Erythema*
   Tarek Hamid, Anush Lingamoorthy, Kyle Quinn, and Amanda Watson
   Proc. IEEE-EMBS Int. Conf. on Body Sensor Networks (IEEE BSN), 2024.

6. *Using Decision Tree Classifier to Increase Screening Test Sensitivity for the Prediction of ACL Retear*
   Tanishik Govil, **Tarek Hamid**, Kimberly Helm, Elliot Greenberg, Kevin Landrum, J. Todd R. Lawrence, Theodore J. Ganley, Amanda Watson
   UBICOMP'24: Adjunct Proceedings of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing, Melbourne, Australia 2024

7. *SpectraVue - An Interactive Web Application Enabling Rapid Data Visualization and Analysis for Wearable Spectroscopy Research*
   Tarek Hamid, Insup Lee, Amanda Watson
   UBICOMP'23: Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing, Cancun, Mexico 2023

8. *Alleviation of Arthritic Symptoms through Thermal Therapy*
   Tarek Hamid, Steven Ayala, Aakash Trivedi, Avi Shah
   In Proceedings of the 2017 Northeast Biomedical Engineering Conference (NEBEC). Newark, NJ: IEEE

9. *(In review) Contact Pressure Estimation using Wearables*
   Tarek Hamid, Patricia Flores, Amanda Watson.

10. *(In review) Glucose Estimation using Wearables*
    Anush Lingamoorthy, Abhishek Murtha, **Tarek Hamid**, Kyle Quinn, Nagarajan Kandasamy, Amanda Watson.

11. *(In review) Lactate Estimation using Wearables*
    Haoran Zhang, Patricia Flores, Michael Clark, Yuxuan Liu, Elizabeth C. Courtney, Xi Chen, **Tarek Hamid**, Robert Theile, Emily Grammar, Kyle C. Quinn, Amanda Watson

INVITED TALKS (EXCLUDES CONFERENCE PRESENTATIONS)

Navigating the Industry as a New Graduate *November 2024*
Duke Industry Series for AI Master of Engineering Students

PATENTS

1. *Objective quantification of melanin, erythema and skin-tone using wearable optical spectroscopy*
   Tarek Hamid, Amanda Watson, Kyle Quinn
   US Patent #10885-083US1

2. *System and Method for Multi-Wavelength Optical Sensing Framework for Calibration-Free Wearable Blood Pressure Monitoring*
   Tarek Hamid, Amanda Watson
   US Patent #10885-090US1

SOFTWARE ARTIFACTS

1. SpectraVue - rapid data visualization and analysis for Lumos
   https://github.com/tarek-hamid/SpectraVue

2. Lumos: An Open-Source Device for Wearable Spectroscopy Research
   https://github.com/LumosIMWUT/Lumos

3. *Raproto: An Open-Source Platform for Rapid Prototyping of Wearable Medical Devices*
   https://github.com/weimerj/Raproto-WearOS
   https://github.com/weimerj/Raproto-Arduino

4. *Digital Biomarker Discovery Pipeline GUI*
   https://github.com/DigitalBiomarkerDiscoveryPipeline/DBDP-GUI

RESEARCH EXPERIENCE

PRECISE Center *2020 - 2022*
University of Pennsylvania

- Developed and integrated IoT-based health monitoring systems using ThingsBoard and VitalCore, to enable real-time physiological monitoring and data processing for devices deployed in clinical trials.
- Designed and developed the Raproto mobile application, a mobile application enabling rapid prototyping of Android smartwatches for data collection and monitoring.
- Performed signal processing and data analysis for the Lumos glucose project, developing pipelines to extract meaningful physiological features from optical sensor data.

Big Ideas Lab *2021 - 2022*
Duke University

- Built an interactive web-based dashboard (DBDP-GUI) that enabled clinicians and researchers to define and explore digital biomarkers from wearables data.

ENTREPRENEURIAL ACTIVITIES

Co-founder *Jan 2026 - present*
OxyVera
Developing the first skin-tone invariant pulse oximeter.

MENTORED STUDENTS

1. **Patricia Flores**, Undergrad Research on *Development of Infrared Wearable Optical Sensors for Biomarker Monitoring* *2024-present*
2. **Tanishk Govil**, Undergrad Research on *Noninvasive Continuous Blood Pressure* *2024-present*
3. **Tanishk Govil**, Undergrad Research on *ACL Re-tear Prediction* *2023-2024*
4. **Xi Chen**, Masters Research on *Robust Pulse Rate Extraction across Skin Tones Using Multi-Wavelength PPG* *2024-2025*
5. **Wilson Beckman**, Post-Undergraduate Mentorship: *Now Machine Learning Data Scientist at Ascension* *2021-2022*
6. **Celine Harris**, TCNJ Engineering Mentorship Program: *Now Senior Associate at Grant Thornton* *2020-2021*
7. **Daniel Young**, Post-Undergraduate Mentorship: *Now MS in Applied Economics at Boston University* *2020-2021*
8. **Harish Erra**, TCNJ Engineering Mentorship Program: *Now Account Executive at Fingerpaint Medical* *2018-2019*

PROFESSIONAL SERVICE AND AFFILIATIONS

Workshop/Conference Organization

1. IEEE International Conference on Healthcare Informatics *2026*
   TRUST-AI 2026: Trustworthy AI for IoMT *Technical Program Committee*

Journal and Conference Reviews

1. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* *2025, 2026, 2027*
2. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics* *2026*
3. *IEEE Body Sensor Networks (BSN)* *2025, 2026*
4. *IEEE Engineering in Medicine and Biology Society (EMBC)* *2024, 2025, 2026*
5. *ACM Health* *2025, 2026*
6. *Elsevier Smart Health* *2024, 2025*

Professional Affiliations

1. *IEEE EMBS: Technical Community on Wearable Biomedical Sensors and Systems (WBSS)*
2. *IEEE*
3. *ACM*
4. *SIGCHI*

TEACHING EXPERIENCE

Teaching Assistant *Fall 2025*
ECE 4501/6501: Sensors and Ubiquitous Computing
Department of Electrical and Computer Engineering, University of Virginia

INDUSTRY EXPERIENCE

VivoSense Jan 2024 - June 2026
Algorithms Engineer *Remote*

Developed algorithms and pipelines to process, analyze, and deliver wearables data from consumer smartwatches to pharmaceutical clients for use in clinical trials.

Sotera Digital Health Jul 2022 - Dec 2023
Biomedical Algorithm Engineer *Remote*

Designed and implemented signal processing and data science algorithms in Python for a next-gen hospital wearable, extracting vital signs such as heart rate, SpO2, and blood pressure from ECG, PPG, and SCG sensor data.

JPMorgan Chase Jun 2020 – July 2022
Software Engineer *New York, NY*

Developed algorithms and internal customer-facing applications to report on the stability of new code changes to the Chase Consumer application using Python, Java, Angular, and TypeScript.

Department of Defense Oct 2019 – June 2020
Electrical Engineer *Picatinny Arsenal, NJ*

Designed custom hardware solutions for military vehicles using Altium.

Johnson & Johnson Jan 2018 – Oct 2019
Scientist *Skillman, NJ*

Led R&D lifecycle management activities for Class I and II consumer medical devices in the North American region.